Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Data Scientist with 3 years of industry experience turning messy data into decisions for non-technical stakeholders, and an MSc in Data Science from the University of Bristol (awaiting result). Analysed customer behaviour with **Market Basket Analysis** and handed the insights to a marketing team, built **Power BI** and **ReactJS** dashboards that moved financial analysts off Excel and saved them **10 hours a week**, and co-built a **calibrated scoring and ranking algorithm** for an MSc dissertation with AstraZeneca, refined with scientist feedback. Evaluated models across multiple seeds with several metrics, analysed **survey data** to predict a hackathon winner (4th place), and worked directly with Compliance, Investment, Client Service and Marketing teams, explaining results and trade-offs in plain language; twice recognised internally with Best Team Project (2024, 2025). Seeking to bring my modelling, analysis and reporting skills to the Labour Party to help turn polling and voter data into campaign and strategy decisions.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Data Science, Statistical Analysis & Modelling

- Analyzed customer data using **Python Pandas** and identified customer buying patterns with **Market Basket Analysis** (**Apriori Algorithm**), handing the insights to the marketing team, who decided how to act on them; used **Pandas** in virtually every project, from data jobs to backend APIs.
- Built a **Python** pipeline on **AWS SageMaker** that turns over **300** daily emails from credit rating agencies into structured data, **fuzzy-matches** company names to the tracked list and sends analysts a consolidated daily summary, **saving 60 manual analyst hours a month**; upgraded parsing from **AWS Textract** to **Azure OpenAI** when a rating agency began sending emails as images, gaining much better accuracy.
- Designed and built **GenAI** applications end to end (**ReactJS**, **Python FastAPI**, **RAG** with **LlamaIndex** and **Azure OpenAI** GPT-3.5 & **Llama-3**) to answer department-specific queries over internal documents, with a separate **pgvector** database per department; first as the only engineer alongside my manager, later leading the team.
- Handled sensitive data responsibly: department-based document access, **role-based access control** with a **Python** authentication function against the company **LDAP** server, and Compliance review of every release; chose models on cost and fit, with **self-hosted Llama-3** for the largest department and stronger **Azure OpenAI** outputs for decision makers.
- Worked directly with business stakeholders in Client Service, Compliance, Investment and Marketing to turn ambiguous requirements into working tools, released them to real users and improved them on their feedback, and explained technical results to non-technical audiences; acted as **de facto team lead** for up to 4 interns in my final year.

Reporting, Dashboards & Data Pipelines

- Developed **Power BI** dashboards for fortnightly credit report analysis for the investment team, migrating financial analysts from **Excel to Power BI** and improving analyst efficiency by **10 hours a week**.

- Recreated the dashboards in **ReactJS** inside the GenAI application, giving the wider organization access to the same data at lower cost than scaling **Power BI** licences.
- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** scripts ingesting the **Morningstar API** for every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly and monthly and integrated with the company's legacy data; developed **Optimized SQL** queries, **SQL Functions** and **Procedures**.

Cloud & Deployment

- Designed the infrastructure architecture for internal analytics platforms and GenAI applications on **AWS** using core services such as **EC2**, **S3**, **ECR** and **Lambda**, with **Docker** microservices.
- Built one-click deployment to dev and prod using **Jenkins** and **GitLab** in a **CI/CD** pipeline, with separate Git repositories and dev/prod branches for frontend and backend.

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built. My focus: data preprocessing across heterogeneous scientific datasets (DepMap, GEO, HPA), the scoring algorithm and the RAG-based AI assistant for a tool that ranks cancer cell lines by target inclusion and exclusion genes; per-gene scoring across six evidence layers, weighted by correlation and calibrated across lineages, with the evidence behind each pick; validated and refined with feedback from AstraZeneca scientists.) | GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (Team project. Multi-phase architecture study comparing CNN/ViT/ResNet encoders and LSTM/attention decoders for generating natural-language image descriptions, evaluated with SBERT cosine, BLEU and ROUGE-L across five seeds and benchmarked against BLIP/SigLIP and commercial VLMs.) | GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **Swindon Borough Council Fuel-Poverty Hack:** household and neighbourhood-level fuel poverty data analysis tool built for the council's hackathon. | GitHub: https://github.com/Arishtanemi4/fuel-poverty.git

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ACHIEVEMENTS

- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git

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SKILLS

- **Tools:** Power BI, Jupyter, Anaconda, AWS (EC2, S3, ECR, Lambda, SageMaker, Textract), Git, GitLab, Jenkins, Docker, Azure OpenAI, LlamaIndex, pgvector, Claude Code, Nginx, Linux, VS Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI), R (dplyr, tidyverse, ggplot2), SQL, NodeJS (express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector), SQLite
